ARTI 308 – Machine Learning

Term 2 – 2025/2026

Project Progress Report

Group #: 4

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Title of the Project: ***Unsupervised Disease Subgroup Discovery from Health Data***

Abstract

Chronic diseases such as hypertension, obesity, and diabetes represent major public health challenges worldwide, yet discovering meaningful patient subgroups from health data without predefined labels remains an open analytical problem. This progress report presents the current state of an unsupervised machine learning framework designed to discover disease subgroups directly from health and dietary indicators, without relying on labeled datasets. The motivation behind this work arises from the increasing availability of health-related data alongside the limitations of traditional supervised analysis methods that depend on predefined disease classifications. Previous studies have primarily focused on disease-specific outcomes using supervised approaches or targeted narrow population groups, leaving a gap in general-purpose, label-free disease subgroup discovery from multi-dimensional health data. To address this gap, the proposed approach employs data preprocessing, feature engineering, and four clustering algorithms: K-Means, Hierarchical Clustering (Ward linkage), DBSCAN, and Gaussian Mixture Models (GMM). Dimensionality reduction via PCA and t-SNE is used for visualization. Preliminary experimental results on a 1,000-record health dataset suggest that GMM achieves the best initial clustering performance among the tested algorithms. However, several important analyses remain incomplete, including feature sensitivity analysis, detailed cluster profiling, and the investigation of patterns between Daily Caloric Intake and other health features. These remaining tasks will be addressed in the next phase of this project.

Keywords: Unsupervised Learning, Disease Subgroup Discovery, K-Means Clustering, Gaussian Mixture Model, Dimensionality Reduction

1. Introduction

Chronic diseases such as hypertension, obesity, and diabetes are among the leading causes of morbidity and mortality worldwide. With the rapid advancement of electronic health records, wearable technologies, and large-scale health surveys, massive volumes of patient health data—including clinical biomarkers, dietary intake, and lifestyle indicators—are continuously being generated. Despite this growth, discovering meaningful disease subgroups from such multi-dimensional data remains a significant analytical challenge, particularly when reliable diagnostic labels are unavailable or incomplete [1][2].

This project introduces an unsupervised disease subgroup discovery framework that applies machine learning techniques to identify latent patient clusters directly from health indicators, without relying on predefined disease labels. The primary aim is to let natural groupings emerge from the data itself, rather than imposing supervised classification boundaries. This approach addresses the limitations of traditional diagnostic models, which require extensive labeled datasets and may introduce bias through predefined assumptions about disease categories [3][4].

Previous studies have explored health data analysis using both statistical and supervised machine learning approaches. Tatoli et al. [1] demonstrated the effectiveness of unsupervised learning in identifying dietary patterns linked to diabetes in Southern Italy. Li et al. [2] applied machine learning to a Japanese cohort to study dietary patterns associated with hypertension. Sun et al. [3] used machine learning to associate dietary patterns with type 2 diabetes risk in elderly Chinese men. Gjoreski et al. [4] explored dietary intake data using unsupervised methods, while Cote et al. [5] used complex network-based algorithms to link dietary behaviors with cardiovascular risks. D'Souza et al. [6] compared various unsupervised methods to understand lifestyle behaviors in children.

While these efforts have provided valuable contributions, they typically depend on labeled datasets and predefined target variables, or focus on disease-specific outcomes within narrow population groups (e.g., elderly, children, or single-gender cohorts). There remains a clear need for exploratory, data-driven approaches capable of discovering disease subgroups from general health data without prior labeling.

The proposed work employs data preprocessing, feature engineering, and four clustering algorithms—K-Means, Hierarchical Clustering (Ward linkage), DBSCAN, and Gaussian Mixture Models (GMM)—along with dimensionality reduction techniques (PCA and t-SNE) to identify meaningful disease subgroups from health indicators. The envisioned outcome is a scalable, label-free analytical framework that can discover clinically interpretable patient clusters and support future research in personalized medicine and public health decision-making.

The remaining part of this work is organized as follows. Section 2 contains a review of related literature. Section 3 describes the proposed machine learning techniques. Section 4 contains the empirical studies including dataset description, experimental setup, and performance measures. Section 5 presents the preliminary results obtained so far and their discussion. Section 6 summarizes the current progress, challenges encountered, and the remaining future work to be completed.

2. Review of Related Literature

Recent advancements in machine learning have enabled researchers to move beyond traditional dietary analysis. Several studies have demonstrated the effectiveness of unsupervised learning in identifying dietary patterns linked to chronic diseases across different global populations.

Tatoli et al. [1] applied unsupervised learning techniques to analyze dietary patterns associated with diabetes in an older population from Southern Italy. Their study identified distinct dietary clusters that correlated with metabolic health markers, demonstrating how clustering methods can reveal diet-disease relationships without predefined labels.

Li et al. [2] investigated dietary patterns associated with hypertension incidence among adult Japanese males using machine learning applied to a cohort study. Their work highlighted how data-driven clustering approaches can identify dietary risk factors in specific demographic groups.

Sun et al. [3] used a machine learning approach to study the association between dietary patterns, obesity, and type 2 diabetes risk in elderly Chinese men. Their findings confirmed the utility of computational methods in uncovering dietary correlations with chronic conditions.

Gjoreski et al. [4] explored dietary intake data collected by food propensity questionnaires using unsupervised learning at an IEEE Big Data conference. This study further highlighted the ability of such methods to uncover hidden patterns without predefined labels.

Cote et al. [5] utilized complex network-based algorithms to link dietary behaviors with cardiovascular risks in the NutriNet-Sante cohort, published in The Journal of Nutrition (2025). Their approach demonstrated that network-derived dietary patterns could identify cardiovascular risk factors.

D'Souza et al. [6] compared children's diet and movement behavior patterns derived from three unsupervised multivariate methods in PLoS ONE (2021). This comparative study provided insights into methodology selection for behavioral pattern analysis.

Table 1: Summary of Related Literature

Ref.	Authors (Year)	Focus Area	Method	Population	Gap
[1]	Tatoli et al. (2022)	Diabetes & dietary patterns	Unsupervised learning	Elderly, Southern Italy	Disease-specific, limited population
[2]	Li et al. (2024)	Hypertension & diet	ML on cohort study	Adult Japanese males	Single-gender, single-disease focus
[3]	Sun et al. (2025)	Obesity, T2D & diet	ML approach	Elderly Chinese men	Narrow demographic group
[4]	Gjoreski et al. (2019)	Dietary intake via FPQ	Unsupervised learning	General	Limited to questionnaire data
[5]	Cote et al. (2025)	CVD risk & diet networks	Network-based ML	NutriNet-Sante cohort	Cardiovascular-specific outcomes
[6]	D'Souza et al. (2021)	Children's diet & movement	3 unsupervised methods	Children	Age-limited scope

Although these studies provide valuable insights, they are often limited in scope and generalizability. Most existing research focuses on disease-specific outcomes such as diabetes or obesity, or targets specific population groups such as the elderly or children. There is still a lack of studies that explore dietary habits in a general population context without relying on predefined health labels or assumptions.

Therefore, a clear research gap exists in applying unsupervised learning techniques to analyze broader and more diverse dietary data. This project aims to address this gap by applying clustering methods such as K-Means along with dimensionality reduction techniques like PCA to uncover latent dietary patterns. The goal is to provide a purely data-driven classification of eating behaviors that can support personalized lifestyle recommendations, rather than focusing solely on clinical diagnosis.

Table 2: Identified Research Gaps

Gap	Present in Literature?
General population analysis (not disease-specific)	✗
Label-free exploratory dietary profiling	✗
Multi-algorithm comparison on same dietary data	✗
Feature sensitivity analysis for dietary biomarkers	✗
Integration of dimensionality reduction with clustering	✓ (partial)

3. Description of the Proposed Techniques

This section describes the four unsupervised learning techniques employed in this project. The selection spans partition-based, hierarchical, density-based, and probabilistic paradigms to provide a comprehensive comparison.

3.1 K-Means Clustering

K-Means is a partition-based clustering algorithm that divides the dataset into K clusters by minimizing the within-cluster sum of squares (inertia). The algorithm iteratively assigns each data point to the nearest centroid and recalculates centroids until convergence. K-Means is computationally efficient and works well with spherical, equally-sized clusters. The optimal number of clusters K is determined using the Elbow Method, Silhouette Score, and Davies-Bouldin Index, supplemented by domain knowledge (three known disease groups). For further details, readers are referred to MacQueen (1967) and Arthur & Vassilvitskii (2007).

3.2 Hierarchical Clustering (Ward Linkage)

Hierarchical Agglomerative Clustering (HAC) builds a tree-like hierarchy of clusters by iteratively merging the two closest clusters. Ward's linkage minimizes the total within-cluster variance at each merge step. Unlike K-Means, hierarchical clustering does not require specifying K in advance; the dendrogram visualization allows inspection of the merging process and selection of an appropriate cut level. This method is particularly useful for understanding the nested structure of dietary behavior groups. For more details, readers are referred to Ward (1963) and Mullner (2011).

3.3 DBSCAN (Density-Based Spatial Clustering)

DBSCAN groups together points that are closely packed in high-density regions, marking outliers in low-density areas as noise. It requires two parameters: eps (neighborhood radius) and min_samples (minimum points to form a dense region). Unlike K-Means, DBSCAN does not require specifying the number of clusters and can discover clusters of arbitrary shape. The eps parameter is selected using the k-distance graph. For further details, readers are referred to Ester et al. (1996).

3.4 Gaussian Mixture Model (GMM)

GMM is a probabilistic soft-clustering model that assumes the data is generated from a mixture of Gaussian distributions. Each data point is assigned a probability of belonging to each cluster, rather than a hard assignment. This makes GMM better suited for overlapping groups, such as patients with comorbid conditions (e.g., obesity and diabetes sharing metabolic features). Model selection uses the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC). For further details, readers are referred to McLachlan & Peel (2000).

4. Empirical Studies

4.1 Description of Dataset

The dataset used in this project is a diet recommendations dataset containing 1,000 patient records with 15 attributes. It includes demographic information (Age, Gender), anthropometric measurements (Weight, Height, BMI), clinical biomarkers (Cholesterol, Blood Pressure, Glucose), dietary indicators (Daily Caloric Intake, Dietary Nutrient Imbalance Score), behavioral measures (Weekly Exercise Hours, Adherence to Diet Plan), and categorical attributes (Disease Type, Preferred Cuisine). The Disease Type attribute contains three classes: Hypertension (398 records), Obesity (393 records), and Diabetes (209 records). The dataset has no missing values.

4.1.1 Statistical Analysis of the Dataset

Table 3: Statistical Analysis of the Dataset

Feature	Mean	Std Dev	Min	25%	50%	75%	Max
Age	49.86	18.11	18.00	35.00	50.00	66.00	79.00
BMI	28.19	8.04	13.00	22.08	27.45	33.42	52.40
Daily Caloric Intake	2475.06	565.02	1500	1984.75	2470.50	2937.25	3498
Cholesterol (mg/dL)	226.10	13.51	200.30	214.38	227.60	237.72	248.90
Blood Pressure (mmHg)	149.26	13.06	125.00	139.00	149.00	159.00	179.00
Glucose (mg/dL)	118.69	25.39	85.10	101.00	111.45	131.18	199.80
Weekly Exercise Hours	5.17	2.85	0.00	2.80	5.20	7.60	10.00
Adherence to Diet Plan	74.88	14.83	50.00	62.00	74.20	88.20	100.00
Nutrient Imbalance Score	2.47	1.46	0.00	1.20	2.40	3.70	5.00

Table 4: Key Correlations Between Features

Feature Pair	Correlation Coefficient
Weight_kg and BMI	0.80
Height_cm and BMI	-0.53
Blood_Pressure_mmHg and Glucose_mg/dL	-0.29
Weight_kg and Glucose_mg/dL	0.32
Blood_Pressure_mmHg and BMI	-0.25
Blood_Pressure_mmHg and Weight_kg	-0.26

The high correlation between Weight_kg and BMI (r = 0.80) justified removing Weight_kg from the feature set to avoid redundancy. Height_cm was also removed due to near-zero correlation with most features.

4.2 Experimental Setup

All experiments were conducted using Python 3.x in Jupyter Notebook / Google Colab. The key libraries used include: Pandas and NumPy for data manipulation, Scikit-Learn for implementing clustering algorithms (K-Means, Agglomerative Clustering, DBSCAN, Gaussian Mixture Model) and dimensionality reduction (PCA, t-SNE), SciPy for hierarchical clustering linkage and dendrogram computation, and Matplotlib/Seaborn for visualization.

The experimental procedure followed these steps:

Step 1 – Data Loading and EDA: The dataset (1,000 records, 15 columns) was loaded and inspected for missing values (none found). Exploratory analysis examined categorical distributions and numerical correlations.

Step 2 – Preprocessing: Patient_ID was dropped as an identifier. Disease_Type was set aside for post-hoc validation only. Gender and Preferred_Cuisine were excluded as they have no medical link to disease groups. Weight_kg was removed due to high correlation with BMI (r = 0.80), and Height_cm was removed due to near-zero correlation with other features. The final feature set consists of 9 numerical features standardized using StandardScaler (zero mean, unit variance).

Step 3 – Dimensionality Reduction: PCA was applied to determine that all 9 components are needed for 95% variance. PCA and t-SNE were used to project data to 2D for visualization.

Step 4 – Clustering: Four algorithms were applied: K-Means (K=2 to 10 sweep, final K=3), Hierarchical Clustering (Ward linkage, K=3), DBSCAN (eps sweep from 40th to 90th percentile of k-distances, min_samples=10), and GMM (3 components, full covariance).

Step 5 – Evaluation: Silhouette Score, Davies-Bouldin Index, and Inertia were computed. Post-hoc validation compared discovered clusters against known Disease_Type labels.

4.3 Performance Measures

Since this project focuses on unsupervised learning, traditional supervised metrics such as accuracy or F1-score are not applicable. The following internal validation metrics are used, consistent with the evaluation strategies found in the literature [1][4][6]:

Silhouette Coefficient: Measures how similar a point is to its own cluster compared to neighboring clusters. Values range from -1 to +1, with higher values indicating better-defined clusters.

Davies-Bouldin Index (DBI): Evaluates the average similarity between each cluster and its most similar one. Lower values indicate better separation and compactness.

Inertia (Within-Cluster Sum of Squares): Measures how tightly data points are grouped around their cluster centroids. Used with the Elbow Method to determine optimal K.

BIC/AIC (for GMM): Bayesian and Akaike Information Criteria measure model fit while penalizing complexity.

Post-hoc Validation: Cross-tabulation of discovered clusters against known Disease_Type labels to assess cluster purity and interpretability.

5. Preliminary Results and Discussion

This section presents the preliminary results obtained so far. The team has completed initial experiments with all four clustering algorithms and conducted an early comparison. These results are subject to further refinement in subsequent phases of the project.

5.1 Exploratory Data Analysis

The dataset contains balanced representation across disease types: Hypertension (398), Obesity (393), and Diabetes (209). Gender distribution and cuisine preferences are also approximately balanced. The correlation heatmap revealed that Weight_kg and BMI share a strong positive correlation ($r = 0.80$), while Blood Pressure and Glucose show a moderate negative correlation ($r = -0.29$).

5.2 Dimensionality Reduction

PCA analysis revealed that all 9 principal components are required to explain 95% of the total variance, indicating that the features carry relatively independent information. The first two components explain only

29.9% of variance, suggesting substantial information exists across multiple dimensions. This motivates the use of t-SNE alongside PCA for 2D visualization.

5.3 K-Means Clustering Results

The K-Means model selection process evaluated K values from 2 to 10. While the Elbow Method did not reveal a sharp elbow, K=3 was selected based on domain knowledge (three disease groups: Hypertension, Obesity, Diabetes). The initial K-Means model (K=3) achieved a Silhouette Score of 0.1071 and a Davies-Bouldin Index of 2.5714 with an Inertia of 7285.74.

5.4 Hierarchical Clustering Results

Hierarchical Agglomerative Clustering with Ward linkage was applied with K=3. A dendrogram was generated from a 200-sample subset for readability. The model achieved a Silhouette Score of 0.0984 and a Davies-Bouldin Index of 2.6234, performing slightly below K-Means on both metrics in this initial evaluation.

5.5 DBSCAN Results

DBSCAN was applied with min_samples=10 and eps values swept from the 40th to 90th percentile of k-distances. Across all tested eps values, DBSCAN consistently found only 1 cluster, with noise points decreasing from 144 (at eps=2.013) to 2 (at eps=2.413). This preliminary finding indicates the data may have uniform density with no natural density-based separations. Further parameter tuning will be explored in the next phase.

5.6 Gaussian Mixture Model Results

GMM with 3 components and full covariance achieved the best initial performance among all algorithms, with a Silhouette Score of 0.1097 and a Davies-Bouldin Index of 2.4997. The BIC was 25089.39 and AIC was 24284.52. The mean maximum probability per point was 0.934, indicating high confidence in cluster assignments. An early post-hoc validation showed that the Hypertension cluster achieved 97.5% purity (390/400), the Diabetes cluster achieved 88.4% purity (152/172), and the Obesity cluster achieved 87.1% purity (373/428).

Table 5: GMM Post-hoc Validation – Discovered Clusters vs. Actual Disease Type

Discovered Group	Diabetes	Hypertension	Obesity	Total
Cluster -> Diabetes	152	0	20	172
Cluster -> Hypertension	10	390	0	400
Cluster -> Obesity	47	8	373	428

5.7 Preliminary Algorithm Comparison

Table 6 below summarizes the initial comparison of all four algorithms. Based on these preliminary results, GMM appears to be the most promising model for this dataset. However, further analysis is needed before drawing definitive conclusions.

Table 6: Preliminary Algorithm Comparison Summary

Algorithm	Clusters	Silhouette Score	Davies-Bouldin Index
K-Means	3	0.1071	2.5714
Hierarchical (Ward)	3	0.0984	2.6234
GMM	**3**	**0.1097**	**2.4997**
DBSCAN	1	N/A	N/A

5.8 Observations So Far

Based on the preliminary experiments conducted, some initial observations can be noted. GMM appears to outperform the other algorithms on this dataset, while DBSCAN does not appear suitable due to the uniform density distribution of the data. However, several important analyses remain to be completed before any firm conclusions can be drawn. In particular, feature sensitivity analysis, detailed cluster profiling, and the investigation of relationships between Daily Caloric Intake and other features such as BMI, Glucose, and Blood Pressure have not yet been fully explored. These remaining analyses are expected to provide deeper insight into the nature of the discovered subgroups and will be addressed in the next phase of the project.

6. Current Progress and Remaining Work

This section summarizes the work completed so far and outlines the tasks that remain for the final phases of the project.

6.1 Summary of Completed Work

To date, the team has accomplished the following milestones:

Data Collection and Preparation: The 1,000-record health dataset was successfully loaded, inspected, and verified to contain no missing values. Exploratory data analysis was completed, including distribution analysis and correlation assessment.

Feature Engineering and Preprocessing: Irrelevant and redundant features (Patient_ID, Gender, Preferred_Cuisine, Weight_kg, Height_cm) were removed based on domain knowledge and correlation analysis. The remaining 9 numerical features were standardized using StandardScaler.

Dimensionality Reduction: PCA and t-SNE were applied for 2D visualization. PCA analysis confirmed that all 9 components are needed to explain 95% of the variance.

Initial Clustering Experiments: All four proposed algorithms (K-Means, Hierarchical Clustering, DBSCAN, GMM) were implemented and evaluated using Silhouette Score and Davies-Bouldin Index. Preliminary results suggest GMM performs best among the four.

6.2 Challenges Encountered

Several challenges have been encountered during this phase of the project:

Low Silhouette Scores: All algorithms produced relatively low Silhouette Scores (below 0.11 with 9 features), indicating substantial overlap between clusters. This suggests the 9-feature space may contain noise that obscures natural groupings.

DBSCAN Unsuitability: DBSCAN consistently produced only a single cluster, indicating that density-based methods may not be appropriate for this particular dataset. This was an unexpected finding that required additional investigation.

Unexplored Dietary Patterns: The relationship between Daily Caloric Intake and other health features (such as BMI, Glucose, Blood Pressure, and Cholesterol) has not yet been thoroughly analyzed. Understanding how caloric intake interacts with clinical biomarkers across the discovered subgroups could reveal important dietary patterns relevant to disease risk.

6.3 Future Work

The following tasks remain to be completed in the next phase of the project:

(1) Conduct a comprehensive feature sensitivity analysis by reducing the feature set to key biomarkers (e.g., BMI, Glucose, Blood Pressure) and comparing clustering performance with the full 9-feature model.

(2) Perform detailed cluster profiling by computing and visualizing the mean feature values for each discovered disease subgroup, including radar charts for clinical interpretation.

(3) Investigate patterns and correlations between Daily Caloric Intake and other features such as BMI, Glucose, Blood Pressure, and Cholesterol across the discovered clusters. This analysis aims to uncover whether caloric intake plays a differentiating role in disease subgroup formation and whether specific dietary patterns are associated with particular disease risks.

(4) Explore ensemble clustering approaches that combine the strengths of K-Means and GMM to potentially achieve more robust subgroup discovery.

(5) Conduct a detailed comparison of our results with the specific studies in the literature review that used similar health datasets, particularly [1] and [3].

(6) Evaluate the potential for deriving personalized health recommendations based on discovered subgroup profiles.

(7) Prepare the final report and presentation with comprehensive documentation of all experimental findings, clinical interpretations, and actionable insights for public health decision-making.

References

[1] R. Tatoli, L. Lampignano, I. Bortone, and R. Donghia, "Dietary patterns associated with diabetes in an older population from Southern Italy using an unsupervised learning approach," Sensors, vol. 22, no. 8, 2022.

[2] L. Li, H. Momma, H. Chen, and Y. Xu, "Dietary patterns associated with the incidence of hypertension among adult Japanese males: application of machine learning to a cohort study," European Journal of Nutrition, 2024.

[3] H. Sun, L. Zhu, P. Wang, and Y. Cui, "Dietary patterns and obesity are associated with type 2 diabetes risk in elderly Chinese men: a machine learning approach," Frontiers in Public Health, vol. 13, 2025.

[4] M. Gjoreski, S. Kochev, and N. Rescic, "Exploring dietary intake data collected by FPQ using unsupervised learning," in Proc. IEEE International Conference on Big Data, 2019.

[5] M. Cote, J. M. Hutchinson, M. Touvier, and B. Srour, "Associations between dietary pattern networks derived from machine learning algorithms and cardiovascular risk in the NutriNet-Sante cohort," The Journal of Nutrition, 2025.

[6] N. J. D'Souza, K. Downing, G. Abbott, and L. Orellana, "A comparison of children's diet and movement behaviour patterns derived from three unsupervised multivariate methods," PLoS ONE, 2021.